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Haemorrhagic Fever

By

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Objective of this lecture is to:

- 1-Explain epidemiology of haemorrhagic fever disease in Iraq
- 2- Identify its occurrence and distribution, clinical features and treatment,
- 3- Understand the program of Control and prevention



→ This is endemic disease in our country that often becomes epidemic

Viral hemorrhagic fevers (VHFs) are a group of diseases that are caused by several distinct families of viruses.

The term “viral hemorrhagic fever” refers to a condition that affects many organ systems of the body, damages the overall cardiovascular system, and reduces the body's ability to function on its own.



Viral hemorrhagic fevers are infectious diseases that can cause severe, life-threatening illness.

They can damage the walls of tiny blood vessels, making them leak, and can hamper the blood's ability to clot.

These diseases most commonly occur in tropical areas.



In the Eastern Mediterranean Region, the main viral haemorrhagic fevers are

Crimean–Congo haemorrhagic fever → only this one in Iraq

Yellow fever,

Dengue fever,

Ebola virus disease.

Rift Valley fever,



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Crimean-Congo Haemorrhagic Fever (CCHF)



Crimean-Congo haemorrhagic fever (CCHF)

Is a widespread disease caused by a tick-borne virus
(Nairovirus) of the Bunyaviridae family.

The CCHF virus causes severe viral haemorrhagic fever outbreaks, with a case fatality rate of 10–40%.



The disease was first characterized in the Crimea in 1944 and given the name Crimean hemorrhagic fever.

It was then later recognized in 1969 as the cause of illness in the Congo, thus resulting in the current name of the disease.

CCHF is endemic in Africa, the Balkans, the Middle East and Asian countries south of the 50th parallel north – the geographical limit of the principal tick vector.



The Crimean-Congo haemorrhagic fever virus in animals and ticks

The hosts of the CCHF virus include a wide range of wild and domestic animals such as cattle, sheep and goats. Many birds are resistant to infection, but ostriches are susceptible



Animals become infected by the bite of infected ticks and the virus remains in their bloodstream for about one week after infection, allowing the tick-animal-tick cycle to continue when another tick bites.

↳ Tick A infect animal A Then Tick virus from Animal A and infect Animal B

Although a number of tick genera are capable of becoming infected with CCHF virus, ticks of the genus *Hyalomma* are the principal vector.



Transmission

The CCHF virus is transmitted to people either by

Tick bites or *↳ But more commonly bite animal*

Through **contact with infected animal blood or tissues** during and immediately after slaughter.

The majority of cases have occurred in people involved in the livestock industry, such as **agricultural workers, slaughterhouse workers and veterinarians.**

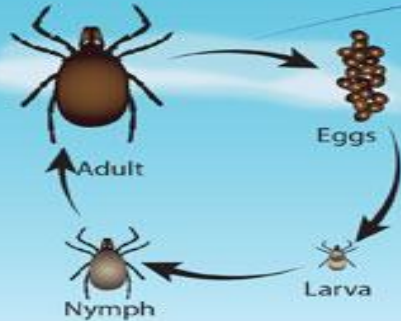


Crimean-Congo Hemorrhagic Fever (CCHF) Virus Ecology

Enzootic Cycle

Ixodid (hard) ticks are both a reservoir and vector for the CCHF virus.

The virus is maintained in nature transovarially and transstadially.



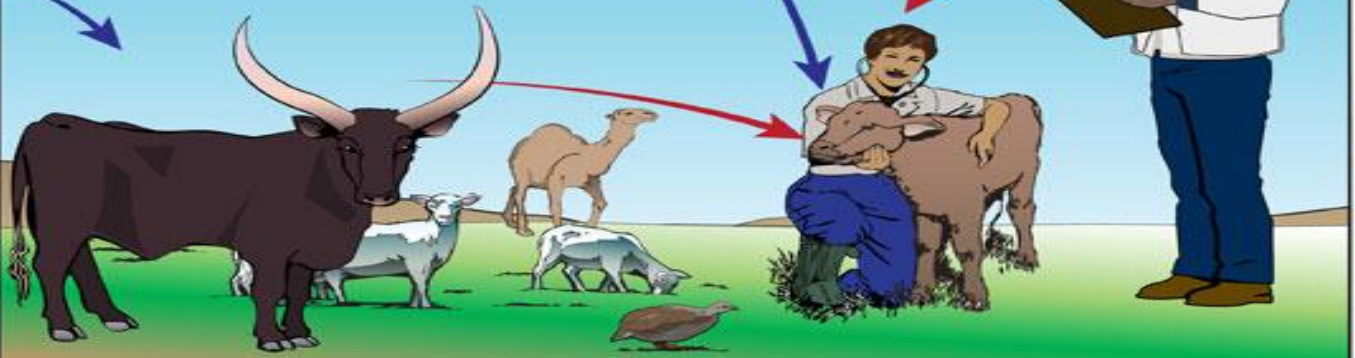
Epizootic-Epidemic Cycle

CCHF cases occur more during the warmer parts of the year, mostly the spring and summer. There are no cases during the winter.

Humans become infected through tick bites and direct contact with infected animal blood or tissue.

Transmission can occur while slaughtering infected animals, during veterinary procedures, and in hospital settings where proper protective equipment and appropriate disinfection procedures are lacking.

Ticks feed on numerous wild and domestic animals such as cattle, goats, sheep, birds, and hares. These animals serve as both food sources for ticks and amplifying hosts for the CCHF virus.





Human-to-human transmission can occur resulting from close contact with the blood, secretions, organs or other bodily fluids of infected persons. *usually if you have like a cut in your hands*

Hospital-acquired infections can also occur due to:

Improper sterilization of medical equipment

Reuse of needles

Contamination of medical supplies.



Signs and symptoms

Following infection by a tick bite, the incubation period is usually one to three days, with a maximum of nine days.

The incubation period following contact with infected blood or tissues is usually five to six days, with a documented maximum of 13 days.



Onset of symptoms is sudden, with fever, myalgia, (muscle ache), dizziness, neck pain and stiffness, backache, headache, sore eyes and photophobia (sensitivity to light).

There may be nausea, vomiting, diarrhoea, abdominal pain and sore throat early on, followed by sharp mood swings and confusion.



After two to four days, the abdominal pain may localize to the upper right quadrant, with detectable hepatomegaly.

Other clinical signs include tachycardia , lymphadenopathy (enlarged lymph nodes), and a petechial rash (a rash caused by bleeding into the skin) on internal mucosal surfaces, such as in the mouth and throat, and on the skin.

Petechial rash + fever + history of contact / exposure to tick or animal = high chance of CCHF
also people in contact with male (especially having cut in hand), places having the ticks
contaminated meat can be cleansed by freezing for 3 days or boiling



The petechiae may give way to larger rashes (ecchymoses) , and other haemorrhagic phenomena.

There is usually evidence of hepatitis, and severely ill patients may experience rapid kidney deterioration, sudden liver failure or pulmonary failure after the fifth day of illness.

Symptoms consist of hematomas



could rarely happen CCIT
meningococcal more likely to
have neck stiffness no ecchymosis Sust
petechia no history of contact







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Diagnosis

CCHF virus infection can be diagnosed by several different laboratory tests:

- . Enzyme-Linked Immunosorbent Assay (ELISA) ;
- . Antigen detection;
- . Serum neutralization;
- . Reverse transcriptase polymerase chain reaction (RT-PCR) assay; and
- . Virus isolation by cell culture.



Treatment

General supportive care with treatment of symptoms is the main approach to managing CCHF in people.

The antiviral drug ribavirin has been used to treat CCHF infection with apparent benefit.

both oral and intravenous formulations seem to be effective.

Prevention and control
Controlling CCHF in
animals and ticks

Prevention and control

Controlling CCHF in animals and ticks





- Tick control with acaricides** (chemicals intended to kill ticks) is only a realistic option for well-managed livestock production facilities.
- Measures were taken to ensure that animals remained tick free for **14 days in a quarantine** station before slaughter.
- There are no vaccines available for use in animals.



Reducing the risk of infection in people

There is currently no safe and effective vaccine widely available for human use.

The way to reduce infection in people is by raising **awareness of the risk factors and educating people** about the measures they can take to reduce exposure to the virus.



Public health advice should focus on several aspects.

• Reducing the risk of tick-to-human transmission:

- wear protective clothing (long sleeves, long trousers);
- wear light colored clothing to allow easy detection of ticks on the clothes *white or yellow*
- use approved acaricides (chemicals intended to kill ticks) on clothing



- use approved repellent on the skin and clothing
- regularly examine clothing and skin for ticks; if found, remove them safely
- seek to eliminate or control tick infestations on animals or in stables and barns
- avoid areas where ticks are abundant and seasons when they are most active.

→ spring and early summer



. Reducing the risk of animal-to-human transmission:

- wear gloves and other protective clothing while handling animals or their tissues in endemic areas, notably during slaughtering, butchering and culling procedures in slaughterhouses or at home;
- quarantine animals before they enter slaughterhouses or routinely treat animals with pesticides two weeks prior to slaughter.



. **Reducing the risk of human-to-human transmission in the community:**

- avoid close physical contact with CCHF-infected people
- wear gloves and protective equipment when taking care of ill people
- wash hands regularly after caring for or visiting ill people.



Controlling infection in health-care settings

Health-care workers caring for patients with suspected or confirmed CCHF, or handling specimens from them, should implement standard infection control precautions.

These include:

basic hand hygiene

use of personal protective equipment

safe injection practices and safe burial practices.



As a precautionary measure, health-care workers caring for patients immediately outside the CCHF outbreak area should also implement standard infection control precautions.

Samples taken from people with suspected CCHF should be handled by trained staff working in suitably equipped laboratories.